Kyle Levy

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Technical Skills

- **Programming:** Python, C, Golang, TypeScript, Assembly, MATLAB, ROS2
- **Embedded Systems:** Arduino, ESP32, FPGA Programming, Embedded Linux
- **DevOps & Cloud:** Git, Kubernetes, Jenkins, Docker, NGINX, S3, OAuth2, OIDC
- **Hardware Design:** SolidWorks, Fusion360, KiCAD, LTSpice, EasyEDA, Gazebo Simulator
- **Testing & Tools:** Oscilloscope, Soldering, 3D Printing, Simulink, Power Tools, Electrical Prototyping

Experience

- **MDA Space** — Brampton, ON
  Student Engineering Intern - Guidance, Navigation and Controls - AI Research — *August 2024 — August 2025*
  - **AI/Machine Learning & Computer Vision**: Spearheaded research, development, and presentation of cutting-edge AI methods for space industry applications, including anomaly detection, visual inspection, black-box optimization, and generative AI. Conducted extensive literature reviews on emerging AI technologies and translated research into practical implementations and potential corporate policy changes regarding the appropriate use of AI autonomy in a high-reliability sector with sensitive data concerns in space applications.
  - **Software Engineering & Backend Development**: Designed, implemented, and documented robust software projects for on-premise cloud infrastructure supporting Large Language Models (LLMs). Architected and integrated secure authentication protocols (OAuth2 & OIDC) and high-redundancy databases to ensure data integrity and accessibility at scale. Delivered a fully featured LLM front-end and infrastructure, incorporating key features for user interaction and data management. Developed and maintained APIs utilizing SQL databases for authentication and access control.
  - **DevOps & Infrastructure**: Established and maintained infrastructure tooling and DevOps processes on-premise. Designed and deployed a multi-node compute cluster that improved AI training speeds by up to 64x, reducing model training time from multiple days to under one hour. Automated CI/CD pipelines using Docker, Jenkins, and Kubernetes to facilitate rapid and reliable deployments. Implemented robust security measures including certificate management, proxy configuration, and authentication protocols.
  - **Robotics & Simulation**: Prototyped and implemented realistic simulations of real-world robotic hardware and interactions, leveraging simulation results to improve physical system performance and inform decision-making. Utilized results to guide the design and optimization of robotic systems. Ran scripts and piloted space grade robots in the lab to run experiments and collect data for AI projects.
  - **Technical Communication & Visualization**: Delivered clear and comprehensive technical documentation for various projects using modern tools (Confluence). Created impactful data visualizations (3D rendering, plotting, PCA analysis) to analyze experimental results and identify correlations between parameters, supporting data-driven decision-making.
  - **Data Parsing**: Engineered custom data pipelines and scripts to parse and process more than 100GB of data from diverse sources, creating a sustainable foundation for AI model training. Explored the implementation of safe agentic workflows for sensitive data with LLM tool calling and forked an open source project to extend its functionality.

- **QMIND (Disruptive Technology Organization)** — Kingston, ON
  Project Manager — *June 2023 — Apr 2024*
  Innovation Design Team Member — *Sept 2022 — Apr 2023*
  - **Leadership**: Led a team of 4 members to build a custom chatbot alternative to be showcased at the CUCAI conference.
  - **Planning & Organization**: Developed multiple statements of work, a timeline for the project, contingency plans, and a method of execution to ensure project success.
  - **Communication and Teamwork**: Coordinated tasks with group members to work effectively and distribute the workload in a way that played to the strengths of each member.
  - **Technical Contributions**: Using Python, assisted with curating data, parsing data, model creation, and model training.
  - **Machine Learning**: Used SKLearn to tokenize the files, build, train, and implement Naive Bayes and Transformer models.
  - **Data Parsing**: Built programs in Python which parsed a dataset of over 250,000 code files that were used to train the model.

Technical Projects

- **Full-Stack Robotics System for Autonomous Subsurface Surveying**: Designed and implemented an autonomous GPR-based mobile surveying system using a Clearpath Husky A100 and Spidar NIC-500. Developed a custom ROS driver enabling real-time GPR data streaming and synchronization. Created a full-stack, Wi-Fi-accessible web dashboard for mission planning, live telemetry, logging, and E-Stop control. Processed and analyzed survey data using unsupervised anomaly detection algorithms to identify subsurface features and geological anomalies.
- **Search and Rescue Rover**: Worked in a pair to develop a semi-autonomous mobile robot which would be used to navigate an environment autonomously and then be manually piloted to manipulate objects remotely. I specialized in the mechanical design of manipulators, electrical systems, wireless communication systems, control systems, and navigation algorithms.
- **Autonomous CO_2 Monitoring System**: Worked with a team to develop an autonomous mobile robot which detected the CO_2 levels in a room for air quality safety purposes. I specialized in using ROS to connect the Raspberry Pi with the Arduino and sensors and the navigation algorithm based on the LiDAR data.

Education

- **Queen's University** Kingston, ON
  Mechatronics and Robotics Engineering, BASc.; GPA: 3.73/4.30 *Sept 2021 — May 2026*
  - **Description:** A new discipline of engineering at Queen's University that combines electrical, mechanical, and computer engineering with a focus on robotics and autonomous systems.
  - **Relevant Courses:** Mechatronics and Robotics Design, Control Systems, Signals and Systems, Sensors and Actuators, Industrial Automation, Robotics, Electronics, Computer Networks, Numerical Methods & Optimization, Embedded Systems & Microprocessors, Computer Architecture, Mobile Robotics, Cryptography & Network Security, Computer Vision & Deep Learning, Advanced Data Analytics, Intelligent Machines & Autonomous Systems, Power Electronics, Biomedical Signals & Image Processing.
  - **Awards:** Queen's University Excellence Scholarship, Dean's Scholar Award (1st, 2nd, & 3rd year)